POAMS: LECTURE 14
THE POPE-OSBORNE ANGULAR MOMENTUM SYNTHESIS

In this lecture, I introduce the Pope-Osborne Angular Momentum Synthesis at last! Our synthesis seeks to explain the phenomenon of natural orbital motion in purely in terms of angular momentum. As a consequence, it is our Normal Realist replacement of Newton's theory of gravitation, as well as other theories. This lecture is based on some of Chapter 6 of 'Light-speed….'.

1. Newton's theory of gravitation

Recall that Newton addressed questions related to falling objects and the orbital motion of the planets by adding to his three laws of motion another law called his *Law of Gravitation* which postulated an *in vacuo* force of attraction between any two bodies. This was formulated as follows:

Every object attracts every other object with a **gravitational force**, **f**, *with magnitude*

$$f = \|\mathbf{f}\| = \frac{GmM}{r^2},$$

where m and M are the masses of the objects concerned, r is the distance between them and G is a constant, known as the **gravitational constant**.

[*As studied in Lecture 13*] The view underlying this assumption was that of the observable universe as a self-sufficient three-dimensional Euclidean space, with time acting as a universal parameter (*i.e.*, not observer dependent), together with a universal invisible *in vacuo* gravitational force which distorted the (assumed) natural rectilinear paths of freely moving objects to form orbital trajectories.

Although the multi-body problem is notoriously intractable, according to Newtonian theory the *total angular momentum* of an *isolated* system of particles is *conserved* if all the forces are *central*, *i.e.*, act along the lines joining pairs of particles. [*End of recap*]

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2. The Pope-Osborne Angular Momentum Synthesis (POAMS)

Our general programme of Normal Realism for physics is to discern the best and most efficient way of interpreting observation with the least amount of theoretical elaboration. This minimalist approach makes much use of 'Ockham's Razor' for economising on unnecessary hypotheses. An example of such an unnecessary hypothesis, in our view, is Newton's idea of an invisible *in vacuo* 'gravitational force'.

Newton's theory of gravitation works well enough on a practical level and is able to produce, amongst other things, Kepler's laws of planetary motion, which are in approximate agreement with observational evidence. However, Newton's theory depends on the existence of an invisible force which can never be directly observed and is therefore unempirical. This in in much the same way as the classical wave theory of light depended on an undetectable *luminiferous ether*. According to Newton's First Law, all *natural* or *free* motion, *i.e.*, motion not subject to any kind of force is rectilinear. However, such rectilinear motion is never observable in practice. All free motion we observe, from satellites to galaxies, is curved. Even apparently rectilinear motion on Earth is curved over a large enough distance, due to the shape of our planet. Newton therefore had to invent the idea of an *in vacuo* gravitational force of attraction to explain why all observable natural motion is curved and not rectilinear.

One of the first natural philosophers to criticize Newton's theory of gravitation was his contemporary, Bishop George Berkley. Berkley argued that to make the vacuum of space a conductor of theoretical 'forces' of any kind was nonsense, and he went so far as to hint that perhaps what was involved might be some direct influence of a universal, *holistic* kind. Much later, Ernst Mach developed this argument of a holistic influence and formulated his well-known principle, which, in essence, states that the influence of distant matter in the observable universe may account not only for gravitational effects, but also for the local phenomena of inertia and momentum.

In Viv and I's approach, Newton's manufactured and artificial *in vacuo* 'force of gravity' is eliminated by simply stating that any continuous *free* or *natural* (inertial) motion is always orbital in the way that the eminent Greek empiricist, Aristotle, observed. Much of Aristotle's original argument is undoubtedly lost in translation, but it went something along

the following lines. Free motion is, by definition, non-interrupted motion. But motion in a straight line, if it is not infinitely long (which was an absurdity to the empirically minded Aristotle) has to begin somewhere and end somewhere. That is to say, it is interrupted, and hence not free. Hence, the only free motion for Aristotle was in a closed cycle, which has no distinguishable initial or final point.

Here, by the *free* or *natural* path of any particle, is meant the path it describes when all restrictions on its motion are removed, that is, when the particle moves freely under the influence of nothing but its own angular momentum. The only 'unnatural' paths in this sense, are those of bodies whose freedom to orbit as they otherwise should, is constrained, such as when a body is in contact with the surface of a planet, or when an artificial satellite is accelerated by a booster rocket. In other words, for 'inertial' of 'force-free' motion, all bodies seek to move, if allowed, in *orbits* with respect to one another, that is, in trajectories of the sort that we observe in astronomical space. In this way, Normal Realism accepts the empirical evidence as it stands that all free motion is orbital and dispenses with the interpretation of 'gravity' as a hidden *in vacuo* mechanism.

Since we claim that we do not require any 'gravitational forces' to explain how it is that freely moving bodies in the observable universe, such as stars and their planets, move in the way that they do, the onus falls on us to describe what agency we propose as an alternative. Newton's mechanistic philosophy prevented him from seriously contemplating any *holistic* or non-local approach to gravitational interaction, of the sort envisaged by Berkley. By contrast, in keeping with our underlying Normal Realist philosophy, our approach, following Berkeley and Mach, begins with the premise that the observable universe is in a holistic balance. Such a balance can only be maintained if the motion of each body in the system is *instantly correlated* with that of each and every other, in conformity with the various holistic conservation laws. Such unmediated instantaneous correlation-at-a-distance is supported by empirical evidence throughout the observable universe. We therefore concur with Tom Phipps, Peter Graneau, Andrew Assis, all participants of the 2001 *Swansea Workshop*, in emphasising an urgent need to rethink our ideas of space, time and motion along more holistic lines. Indeed, in his book, *The Science of Mechanics*, Mach states,

…. every single body of the universe stands in some definite relation with every other body in the universe; … any one body, and consequently also several bodies, cannot be regarded as wholly isolated… We cannot regard it as impossible that integral laws, to use an expression of C. Neumann, will someday take the place of the laws of mathematical elements or differential laws, that now make up the science of mechanics, and that we shall have direct knowledge of the dependence on one another of the positions of bodies. In such an event, the concept of force will have become superfluous.

Accordingly, in our approach, we choose *holistic conservation of angular momentum* as the agency whereby all natural (force-free) motions are instantaneously correlated. Angular momentum is clearly the measure to use in orbital motion, since it contains observational information about the fundamental empirical measures of spatial relations, mass, length and time, as well as orientation. In addition, if we treat the observable universe as an isolated system, then Newtonian theory dictates that, with the assumption of central forces, angular momentum is holistically conserved. This is also the case in the Copenhagen interpretation of Quantum Mechanics [*see Lectures 6 and 11*]. More recently, work by other authors, notably Philip Kanarev, have emphasised the central role of conservation of angular momentum, both in the effect of spin on macroscopic orbital trajectories, and its quantisation in atomic structure. Both of these aspects will be discussed in later lectures.

Another prime candidate for the instantaneously correlating agency for natural motion is holistic conservation of (total) energy. However, angular momentum, being a vector rather than a scalar quantity, provides more information than energy for orbital motion.

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3. <u>Felt and seen distant actions</u>

Let us now consider the notion of *inertia*. This, according to Mach, is a *felt* connection. As he is reputed to have said, ‘When the subway train jerks it is the fixed stars that throw you down’. Changes in angular momentum imply, of course, changes in *angular inertia*. For this reason, these angular momentum connections may be described as *felt* connections. However, just as in the case of any *seen* connection, we have to square the fact that

these immediately felt (for example visceral or sensational) connections are experienced by us instantly with the fact that in observational space, their sources are separated from us by the distance-time factor *c*.

Now a space traveller orbiting freely around Earth does not feel the presence of Earth or the other bodies in and beyond the solar system whose existence and movements, in the whole angular momentum nexus, define his orbital trajectory. It is only when he either accelerates or decelerates that he is immediately aware, by the sensations he feels as soon as he takes the action, that the previous correlation between the motion of his own body and those other bodies has changed. This, of course is what is commonly called 'inertia', except that his inertial trajectory is no longer rectilinear, as in Newtonian theory, but curvilinear. However, while the correlation between the space-traveller's motion-changing action and its accompanying visceral sensations are proper time-instantaneous and his angular momentum connections with those bodies are also proper time-instantaneous, any distant reciprocal reaction will not be detected until the lapse of time *s*/*c*, as explained in Section 5 of Lecture 11.

According to our angular momentum synthesis then, any immediately felt connections, like seen connections, are *proper time* instantaneous, although not *observationally* instantaneous, for those reasons described in previous lectures. In other words, the felt and the seen, are all part of the same quantum-informational manifold. Hence, there is no 'EPR Paradox' associated with the instantaneously felt connections any more than there is with the seen. These felt interactions are of the Lewisian sort described in earlier lectures [*see Lectures 6 and 11*]. That is to say, the agent and reagent particles are physically touching, so there is no question of any vacuum having to be bridged in that interaction. In Normal Realism, then, these interactions are conveyed by the most easily understood sort of action transfer, namely *contiguity* (the state of being in contact).

It should be noted that according to our Normal Realist philosophy, it is pointless trying to find some 'mechanical-causal' reason for how a change in angular momentum in one orbit system, produces a change in angular momentum in another system somewhere else. This is like trying to find 'mechanical-casual' reasons for the fact that in thermal motion, the energies of the particles distribute themselves inexorably in the direction of maximum entropy applied to the system as a whole, as stated in the *Second Law of Thermodynamics*. So far as Normal Realism is concerned,

all of physics is based, not on classical mechanical-causal determinism among particles but on fundamental indeterminacy, governed ultimately by purely statistical laws like that of the Second Law of Thermodynamics, applied holistically.

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4. The fundamental postulates of the Pope-Osborne Angular Momentum Synthesis (POAMS)

In summary then, the fundamental postulates of the *Pope-Osborne Angular Momentum Synthesis,* known by the acronym POAMS, which was first coined by John Blay, are as follows:

Postulate 1: *Natural (inertial) motion is always orbital, the natural path of any particle being the trajectory followed by the particle when it moves freely under the influence of nothing but its own angular momentum (angular inertia).*

Postulate 2: *Angular momentum (angular inertia) is holistically conserved.*

These postulates can be combined in the following principle:

The Principle of Angular Inertia: *All bodies move angular-inertially. Angular inertia is the resistance of a body to any change in its angular momentum relations with other bodies and is the natural tendency of a body to follow the path of least resistance (i.e. least action) in accordance with the conservation of angular momentum.*

Our approach to orbital motion takes the 'force of gravity' out of metaphysical inscrutability and puts it where it truly belongs – that is, in actual physical reality – as the *felt* and *measurable* force required to prevent bodies from travelling in their other naturally force-free (angular inertial) orbits. The POAMS premise is that it is angular momentum which keeps bodies apart, and if angular momentum vanishes, then bodies automatically fall together, in what orthodox physics describes as 'gravitational attraction'.

One of the implications of POAMS is that all talk of 'attraction' and 'repulsion' between freely moving particles becomes redundant. There are only *degrees* of *relative* attraction and repulsion in the sense that a particle P_1 orbiting a body B at a mean radius r_1 can be considered more 'attracted' to B than a particle P_2 orbiting B at a mean radius $r_2 > r_1$.

It is important to realise that, in complete contrast to Newtonian theory, POAMS makes no distinction between the observable universe at the macro- and micro-levels. Moreover, whereas Newtonian theory postulates an underlying continuum, in POAMS, the structure of the 'universe' at the microphysical level is ultimately *discrete* with angular momentum *quantised*, as discussed in previous lectures. As pointed out by Peter Rowlands in the 2001 *Swansea Workshop*, at the quantum level, conservation of angular momentum implies that the three 'charges' responsible for the electromagnetic, strong and weak interactions are separately conserved, so that *angular momentum is essentially a representation of charge*. I shall return to this point in a later lecture. Energy may be regarded as a constituent of angular momentum at the quantum level, but clearly not *vice-versa*, and this is another reason why POAMS considers conservation of angular momentum more fundamental than conservation of energy.

Of course, POAMS is not the only approach to orbital motion which dispenses with *in vacuo* forces. Another, such theory is Einstein's *General Relativity*, which describes space-time by means of a mathematical model known as a *Riemannian manifold*. In General Relativity, it is postulated that naturally orbiting bodies move in the way they do, not because of the existence of *in vacuo* forces, but because they follow the *geodesics*, *i.e.* the 'straightest' paths (paths of least resistance) in an underlying *curved* space-time continuum. In other words, the agency for orbital motion is the 'curvature of the space-time manifold'. It must be remembered that here however, that this manifold is no more than a convenient mathematical structure and is not intended to be a model of actual physical reality. A global space-time continuum which predicts properties of the observable universe as a whole *i.e. cosmological* properties, can be contemplated in General Relativity, and this represents a type of holism. But there is certainly no instantaneous unmediated correlation-at-a-distance in General Relativity, which is bound by Special Relativity, according to which all physical interaction and information are delayed due the universal 'speed

limit' *c*. Moreover, there is no ultimate discreteness contemplated of the sort that POAMS embraces.

In must be emphasised here that POAMS is essentially a *synthesis* of apparently disparate physical theories. It needs to be stressed that it is not a new *theory* intended to replace, for example, Newtonian dynamics, Special and General Relativity or Quantum Mechanics. Indeed, POAMS ultimately seeks to modify these theories to an extent that is no more than necessary to remove the logical contradictions between them and allow them to fit naturally together. In this way, whilst retaining some of the mathematical elements of existing theories, it dispenses with other elements regarded as unnecessary and, in some cases, provides a very different interpretation that is logically coherent and consistent with our Normal Realist philosophy.

Of course, the mere fact that angular momentum is universally conserved cannot be expected to supply, in itself, the reason why bodies orbit in the way that they do. For example, no conservation principle automatically provides an *equation of motion*. On the other hand, equations of motion can provide conservation principles. For example, as demonstrated in the last lecture, in Newtonian gravitation, the existence of centrally directed forces in any isolated system ensures conservation of angular momentum. These forces do not have to be of an inverse-square kind, so that Newton's inverse square law is certainly not a necessary consequence of conservation of angular momentum. However, supposing only that angular momentum is conserved provides more flexibility and produces not only the Newtonian equations of motion as a special case, as demonstrated in the next lecture, but also, by incorporating relativistic effects, provides General Relativistic equations of motion, as will be shown in later lectures.

We now turn our attention to the treatment of *constrained* ('unnatural') trajectories in POAMS. Such trajectories represent a state of motion in which a body is held, by a *real measurable* force, against its natural tendency to orbit freely. In our alternative interpretation, the true (*i.e.* measurable) force is not *concealed* in the natural orbit of the body, but is *revealed* in the difference between that natural orbit and a constrained one. In effect, Newton's Second Law, which defines *force* as the rate of change of linear momentum, is replaced, here, by *torque*, defined as the rate of

change of angular momentum. For example, in POAMS, the weight with which a body presses on the surface of Earth is not the result of some unseen and mysterious ‘pull of gravity’. It is because the body, where it stands on the rotating Earth, has insufficient angular momentum to orbit freely (weightlessly), so that with the angular momentum it has, the *real* force it exerts on a weighing scale is that of the reaction of Earth’s surface preventing it from orbiting where it should, much closer to Earth’s centre.

In the next lecture, I will apply the central postulates of POAMS in our treatment of the two-body problem. This is in a way which does not involve Newton’s in vacuo ‘gravitational force’ postulate.

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